

Orthogonal Sparsity Pressure: Status Update

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1 Overview: Model Architecture and Dataset

Current Model

The current experimental target is the Pythia-14M architecture from EleutherAI/[pythia-14m-deduped](#). In this repository, “Pythia-14M” means the architecture/configuration only: runs initialize weights randomly and do not load released Pythia checkpoint weights. The implementation constructs the model from the Hugging Face config and trains it as a pretraining run.

Table 1: Current Pythia-14M architecture used by the harness.

Item	Value
Transformer blocks	6
Residual hidden width	128
Attention heads	4
Per-head width	32
MLP hidden width	512
Context length	2048
Vocabulary size	50,304
MLP activation	GELU
Positional encoding	RoPE
Attention / hidden dropout	0.0 / 0.0
Trainable parameters	14,067,712
Final checkpoint size	~53.67 MB
Parameter dtype	FP32 parameters; bf16 autocast compute on CUDA

Pythia is an EleutherAI family of GPT-NeoX causal language models designed for controlled research on language-model training dynamics. The original Pythia paper introduces a suite of models trained on public data in the same order, with dense checkpoint coverage and tools for reconstructing training dataloaders, specifically to make analyses across training and scaling more auditable ([Biderman et al., 2023](#)). That makes the family a strong substrate for activation-pressure experiments: the research question depends on model-internal quantities and early pretraining behavior, not only on final downstream task scores.

Pythia-14M is the smallest local starting point in this design. The [Pythia-14M-deduped model card](#) notes that 14M was added after the original 70M–12B suite for interpretability researchers who wanted a smaller model with the same training setup conventions. In this report, 14M is used as an architecture-level testbed with random initialization: it is small enough to run repeated pretraining ablations locally, but remains tied to the Pythia/GPT-NeoX architecture family and therefore preserves a clear path to later 31M, 70M, and 160M scale-ladder checks.

Current Pressure Site

The current activation pressure site is `mlp_hiddens`. For each Pythia block, the harness hooks:

`gpt_neox.layers.{l}.mlp.act`

This captures the post-GELU MLP hidden activation:

$$A_l = \text{GELU}(Z_l) \in \mathbb{R}^{B \times T \times 512},$$

for each layer $l = 0, \dots, 5$. Pressures and near-zero activation measurements are applied to these activation tensors, not to model weights.

Elementwise interpretation. Current L1 and Ricker pressures score scalar activation entries $A_l[b, t, j]$ and then average those scores into one pressure loss. Likewise, reported near-zero mass and post-hoc exact-zero sparsity are elementwise activation statistics across layers, batch items, token positions, and MLP hidden channels. For example, 80% exact-zero activation sparsity does not mean that a fixed ~ 410 of the 512 hidden channels are zero for every batch item and every token position. That stronger claim would require a separate structured channel-sparsity metric or pressure.

Pythia Scaling Ladder View

The current runs are intentionally anchored at 14M. A later scale ladder should only be added after the 14M path is stable enough to justify the added runtime and comparison burden. The table below is a planning view of open-source Pythia-family architecture sizes, not a commitment that all scales will be run locally.

Table 2: Candidate Pythia architecture scale ladder. Approximate weight storage assumes FP32 weights at 4 bytes/parameter and ignores optimizer state.

Scale	Nominal params	Approx. FP32 weights	Status / role
Pythia-14M	14M	~ 54 MB	Current measured local baseline and pressure screen.
Pythia-31M	31M	~ 118 MB	Optional intermediate sanity scale.
Pythia-70M	70M	~ 267 MB	Small scale-up once 14M candidates are selected.
Pythia-160M	160M	~ 610 MB	Near-term upper local target mentioned for this project.
Pythia-410M	410M	~ 1.5 GB	Likely requires separate runtime/memory planning.
Pythia-1B	1B	~ 3.7 GB	Beyond current local screening plan.
Pythia-1.4B	1.4B	~ 5.2 GB	Beyond current local screening plan.
Pythia-2.8B	2.8B	~ 10.4 GB	Probably not suitable for this laptop workflow without major changes.
Pythia-6.9B	6.9B	~ 25.7 GB	Out of scope for current local experiments.
Pythia-12B	12B	~ 44.7 GB	Out of scope for current local experiments.

Source notes for model identities: [EleutherAI Pythia repository](#) and the [Pythia-14M Hugging Face repository](#).

Dataset

The dataset is MiniPile via [JeanKaddour/minipile](#). The repository uses a local tokenized cache rather than streaming during training, so token counts and ordering are auditable.

Table 3: Local tokenized MiniPile cache currently used by fixed-step experiments.

Split	Documents	Tokens	Notes
Train	1,000,000	1,491,711,416	Full local train token cache.
Validation	500	693,668	Local validation subset used for periodic validation.

Tokenizer: [EleutherAI/pythia-14m-deduped](#). Block size: 2048. EOS is appended during tokenization. Dataset source note: [JeanKaddour/minipile](#).

Method Families Under Test

All methods train the same random-initialized Pythia-14M architecture on the same tokenized MiniPile cache. The common task loss is the causal language-modeling loss L_{task} . Activation-pressure methods operate on the captured post-GELU MLP hidden activations $A_l[b, t, j]$ at `mlp_hiddens`. The baseline is AdamW with no activation pressure; pressure variants either add an auxiliary activation loss directly to the training objective or apply a separate Adam-step-space orthogonal pressure correction.

Table 4: Method families in the current activation-pressure screen.

Label	Method	Parameters and current use
AdamW	Baseline optimizer	No activation pressure: <code>method: none, weight: 0</code> . AdamW optimizes L_{task} only; activation metrics are monitor-only. Shared learning rate is 3.0×10^{-5} .
RN	Naive Ricker	Direct augmented-loss pressure: $L_{\text{task}} + wP_{\text{Ricker}}(A)$. Tuned parameters are pressure weight w and Ricker shape parameters c, σ . Current grid varies w over $\{0.03, 0.1, 0.3, 1.0\}$ with selected narrow/wide (c, σ) settings.
OR	Orthogonal Ricker	Same Ricker pressure and parameter grid as RN, but applied as a separate Adam-step-space correction after the task AdamW step. Fixed orthogonal controls: <code>step_budget: 0.5, eps: 1e-12</code> .
L1N	Naive L1	Direct augmented-loss pressure: $L_{\text{task}} + wP_{\text{L1}}(A)$. Tuned parameter is pressure weight w . Current grid uses $w \in \{0.05, 0.15, 0.5, 1.0, 2.0, 5.0\}$.
OL1	Orthogonal L1	Same L1 pressure and weight grid as L1N, but applied as the separate projected/capped Adam-step-space correction. Fixed orthogonal controls: <code>step_budget: 0.5, eps: 1e-12</code> .

The pressure objectives are:

$$P_{\text{L1}}(A) = \text{mean}_{l,b,t,j} |A_l[b, t, j]|$$

$$r(a; c, \sigma) = \left(1 - \frac{a^2}{c^2}\right) \exp\left(-\frac{a^2}{2\sigma^2}\right), \quad P_{\text{Ricker}}(A) = 1 - \text{mean}_{l,b,t,j} r(A_l[b, t, j]; c, \sigma).$$

The “naive” labels mean direct augmented-loss pressure: the pressure gradient participates in the same backward pass as the task loss. The “orthogonal” labels mean the task update and pressure correction are separated; optimizer moments are driven by task gradients, while the pressure correction is applied after the AdamW task step. All pressure objectives are elementwise over activation scalars and then averaged; they do not directly impose structured channel sparsity.

2 Early Calibration and Tests

Fixed-step Run Budget

The fixed-step pressure screen uses one shared budget across AdamW, naive pressure methods, and orthogonal pressure methods.

Table 5: Fixed-step budget used by configs 12–48.

Item	Value
Optimizer steps	2,048
Block size	2,048
Micro-batch size	4
Gradient accumulation	8
Tokens per optimizer step	65,536
Tokens per run	134,217,728
Fraction of MiniPile train token cache	~8.998%
Validation frequency	every 250 steps, plus step 1 and final step
Validation size per evaluation	8 batches = 65,536 tokens

Observed Local Runtime

The table below summarizes observed runtime across completed fixed-step runs on the local RTX 5070 Ti Laptop GPU. Values are measured from saved `metrics.json` files, not projected from theoretical FLOPs.

Table 6: Observed runtime and GPU memory by method family.

Method	Runs	Median wall	Wall range	Median tok/s	Peak alloc.	Peak reserved
AdamW	1	16.1 min	16.1–16.1 min	138,567	5,996 MB	7,430 MB
RN	12	31.1 min	30.8–31.7 min	71,904	6,367 MB	7,440 MB
OR	12	21.7 min	21.4–22.4 min	103,135	6,934 MB	7,446 MB
L1N	6	28.7 min	28.7–28.9 min	77,834	6,175 MB	7,446 MB
OL1	6	20.5 min	20.3–20.5 min	109,329	6,934 MB	7,446 MB

Approximate measured time for one full pass over the 1.49B-token MiniPile train cache at the observed median throughput:

Table 7: Approximate one-pass wall-clock estimates at measured median throughput.

Method	Estimated wall-clock for one token-cache pass
AdamW	~3.0 h
RN	~5.8 h
OR	~4.0 h
L1N	~5.3 h
OL1	~3.8 h

3 Covered Parameter Grid

Ricker Methods

RN denotes naive Ricker pressure in the augmented loss. OR denotes Adam-step orthogonal Ricker pressure. Orthogonal runs use `step_budget: 0.5`.

Table 8: Covered Ricker parameter grid.

Regime	Parameters	RN config	OR config
Initial screen	<code>w=0.03, c=0.05, s=0.05</code>	13	20
Initial screen	<code>w=0.1, c=0.05, s=0.05</code>	14	21
Initial screen and high-pressure overlap	<code>w=0.3, c=0.05, s=0.05</code>	15	22
Initial screen	<code>w=0.1, c=0.02, s=0.02</code>	16	23

Regime	Parameters	RN config	OR config
Initial screen	$w=0.1, c=0.1, s=0.1$	17	24
Initial screen	$w=0.1, c=0.05, s=0.025$	18	25
Initial screen	$w=0.1, c=0.05, s=0.1$	19	26
High-pressure/wide-Ricker expansion	$w=0.3, c=0.1, s=0.1$	35	40
High-pressure/wide-Ricker expansion	$w=0.3, c=0.5, s=0.5$	36	41
High-pressure/wide-Ricker expansion	$w=1.0, c=0.05, s=0.05$	37	42
High-pressure/wide-Ricker expansion	$w=1.0, c=0.1, s=0.1$	38	43
High-pressure/wide-Ricker expansion	$w=1.0, c=0.5, s=0.5$	39	44

L1 Methods

L1N denotes naive L1 activation pressure in the augmented loss. OL1 denotes Adam-step orthogonal L1 pressure. Orthogonal runs use `step_budget:` 0.5.

Table 9: Covered L1 parameter grid.

Regime	Parameters	L1N config	OL1 config
Initial screen	$w=0.05$	27	31
Initial screen	$w=0.15$	28	32
Initial screen	$w=0.5$	29	33
Initial screen	$w=1.0$	30	34
High-pressure L1 expansion	$w=2.0$	45	47
High-pressure L1 expansion	$w=5.0$	46	48

4 Early Experimental Results

Selected Methods

The table below mirrors the selected post-hoc clipping frontier in Figure 1. $\text{Val@}x$ is the validation loss at the nearest achieved exact-zero activation sparsity to x .

Table 10: Selected early fixed-step results.

Method	Parameters	Final val	Tokens/s	NZ ≤ 0.01	NZ ≤ 0.03	Val@0.4	Val@0.6	Val@0.8
AdamW	monitor only	7.02	138.6k	6.5%	19.4%	7.11@0.47	7.21@0.60	7.50@0.81
L1N	$w=0.5$	7.03	77.9k	20.9%	53.7%	7.07@0.23	7.12@0.58	7.33@0.77
OL1	$w=0.5$	7.02	109.4k	21.9%	55.2%	7.06@0.24	7.12@0.59	7.35@0.78
RN	$w0.1 \ c0.05 \ s0.025$	7.08	71.5k	44.4%	65.9%	7.12@0.48	7.14@0.67	7.24@0.78
OR	$w0.1 \ c0.05 \ s0.025$	7.05	103.0k	39.1%	62.3%	7.09@0.43	7.11@0.64	7.37@0.82

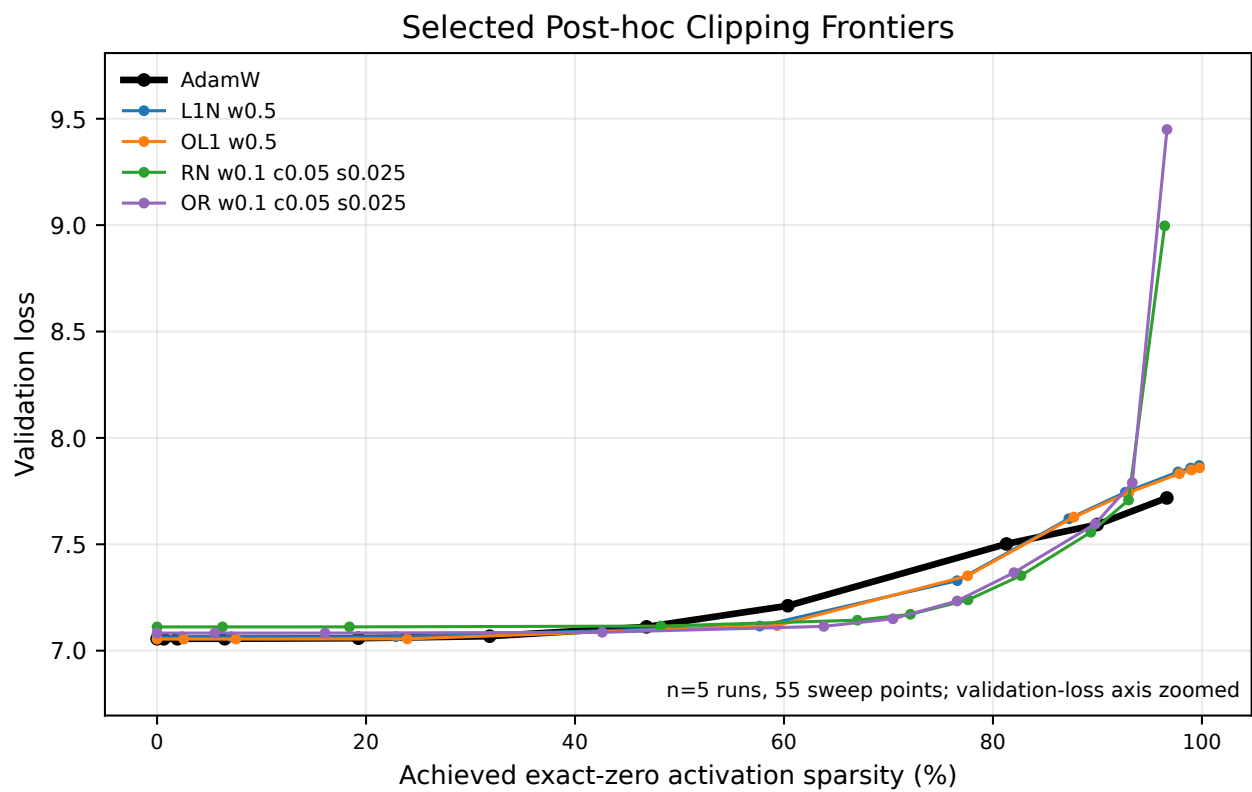


Figure 1: Selected post-hoc clipping frontier for AdamW, L1N, OL1, RN, and OR.

High-pressure RN Learning Curves

n=6 runs: AdamW plus RN configs 35-39; 134,217,728 tokens/run

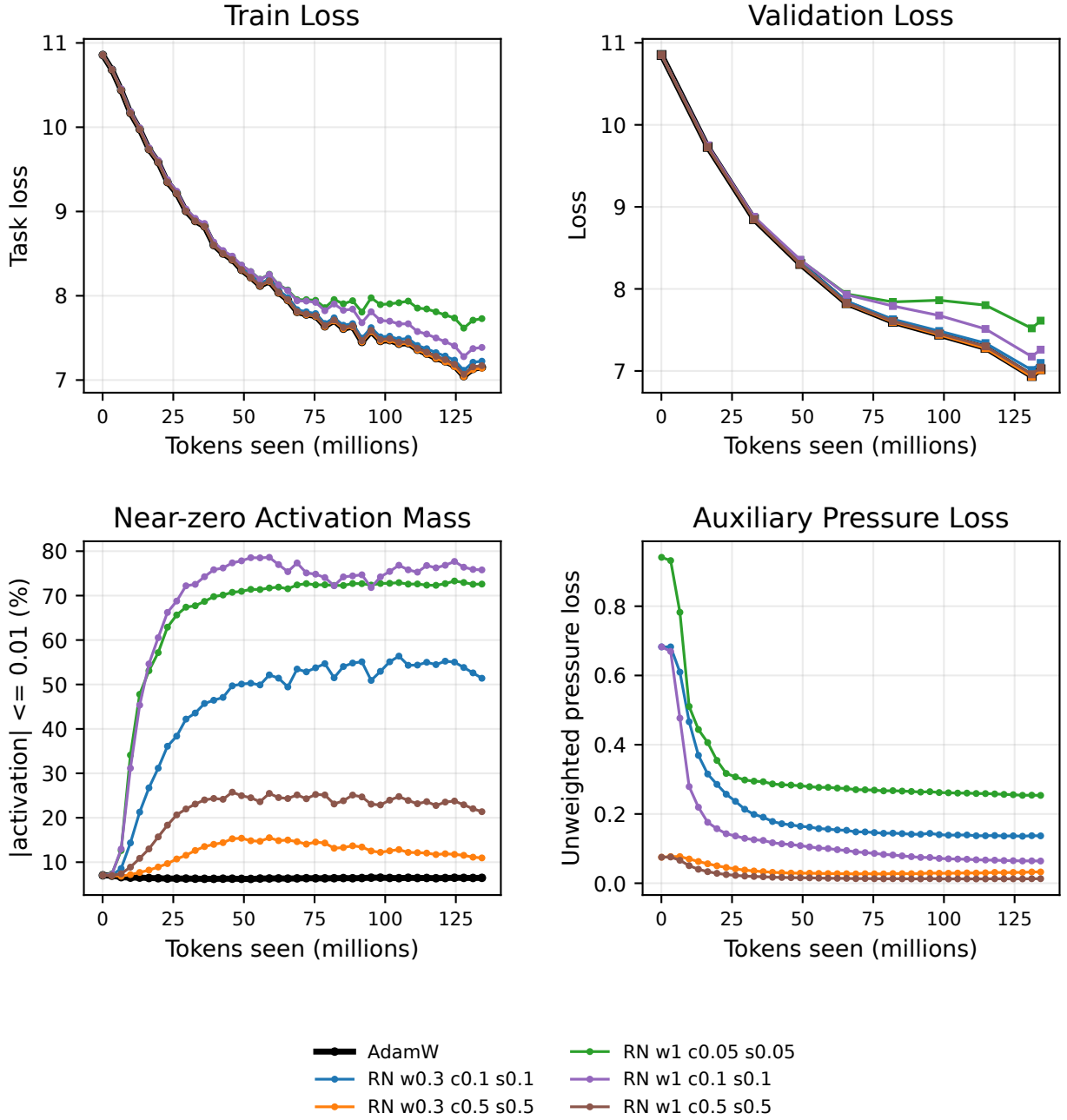


Figure 2: High-pressure RN learning curves, compared against AdamW.

High-pressure OR Learning Curves

n=6 runs: AdamW plus OR configs 40-44; 134,217,728 tokens/run

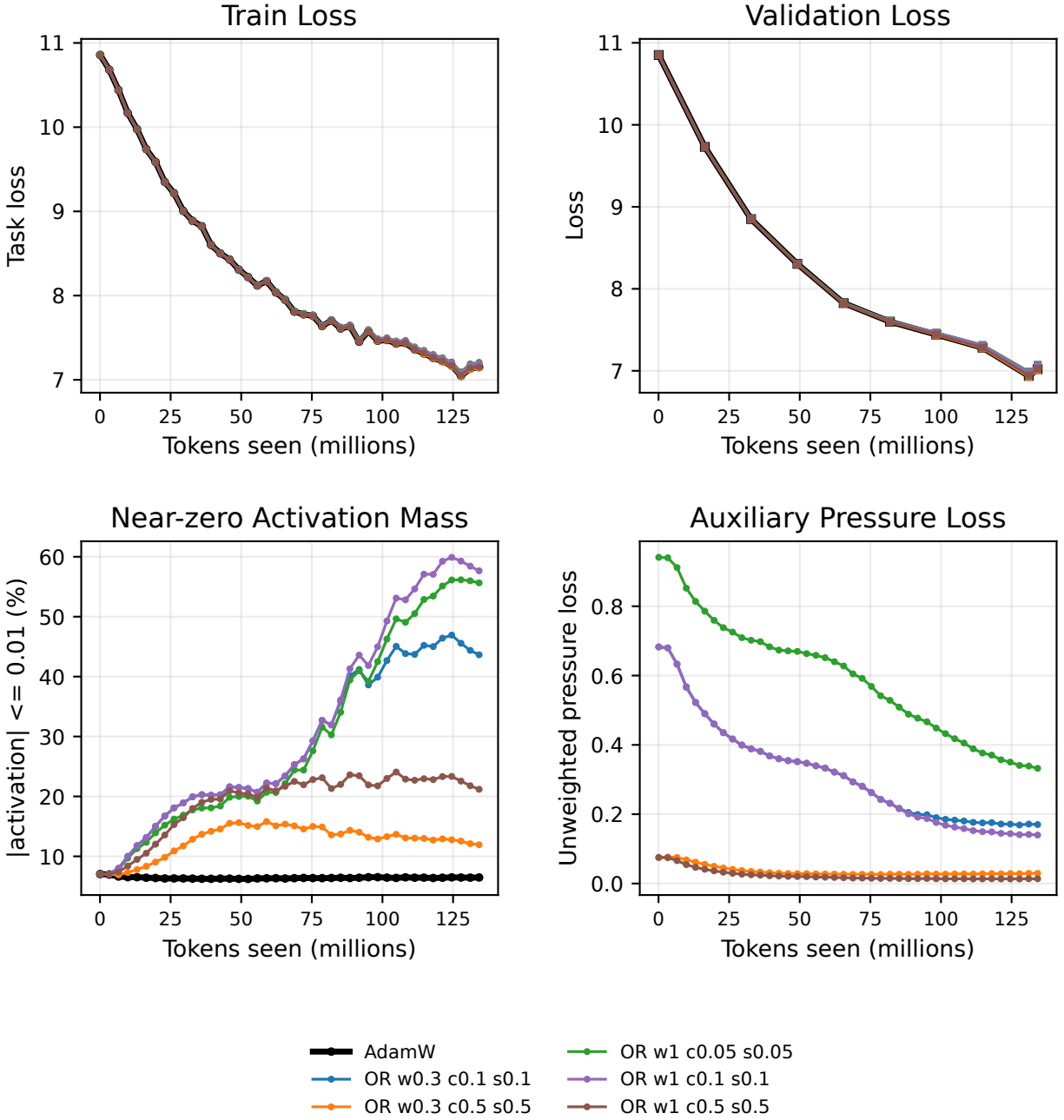


Figure 3: High-pressure OR learning curves, compared against AdamW.

High-pressure L1N/OL1 Learning Curves

n=5 runs: AdamW plus L1N/OL1 configs 45-48; 134,217,728 tokens/run

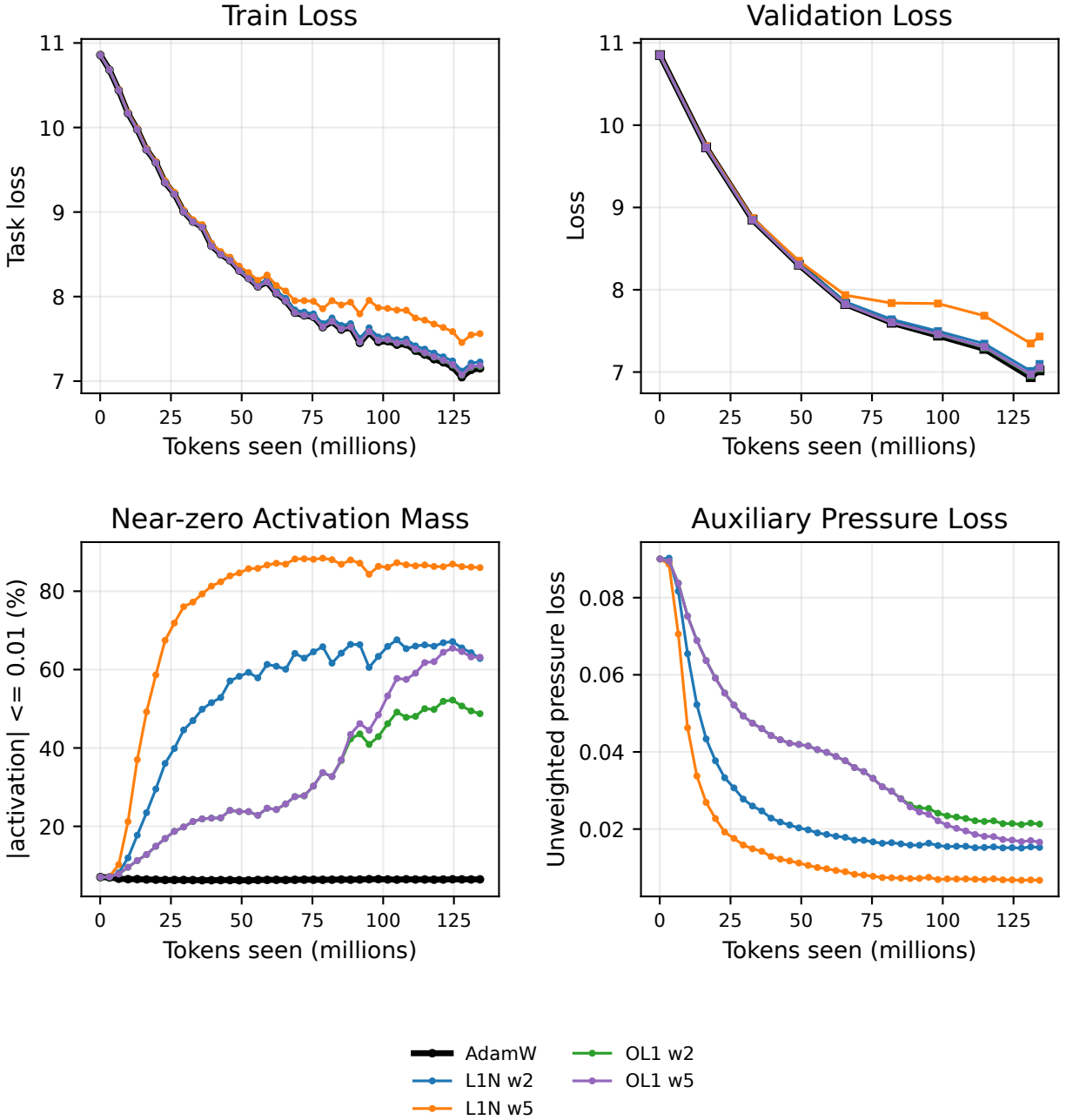


Figure 4: High-pressure L1N and OL1 learning curves, compared against AdamW.

High-pressure OR and L1 Weight Norm Diagnostics

n=10 runs: AdamW, OR configs 40-44, and L1N/OL1 configs 45-48; up to 134,217,728 tokens/run

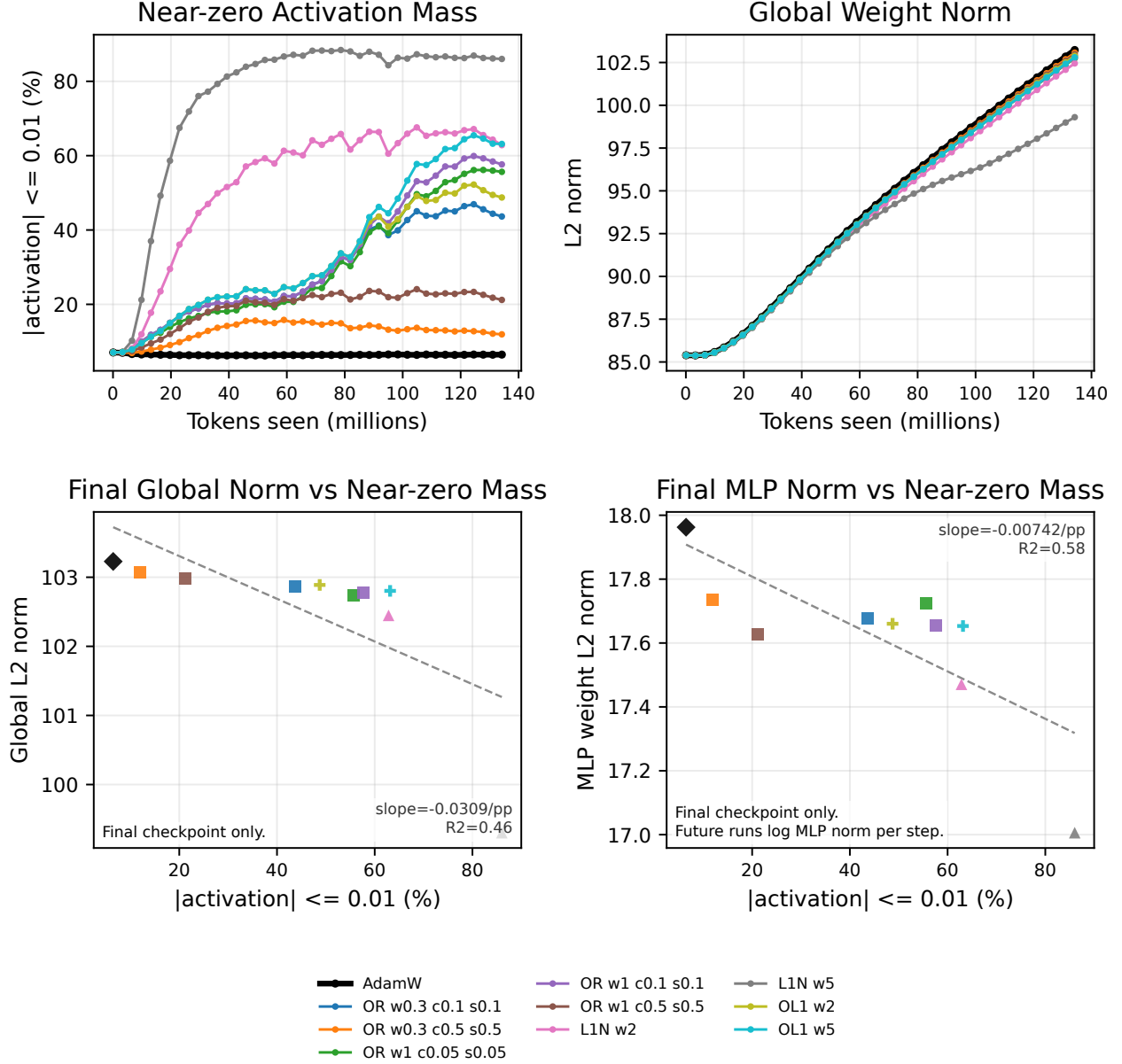


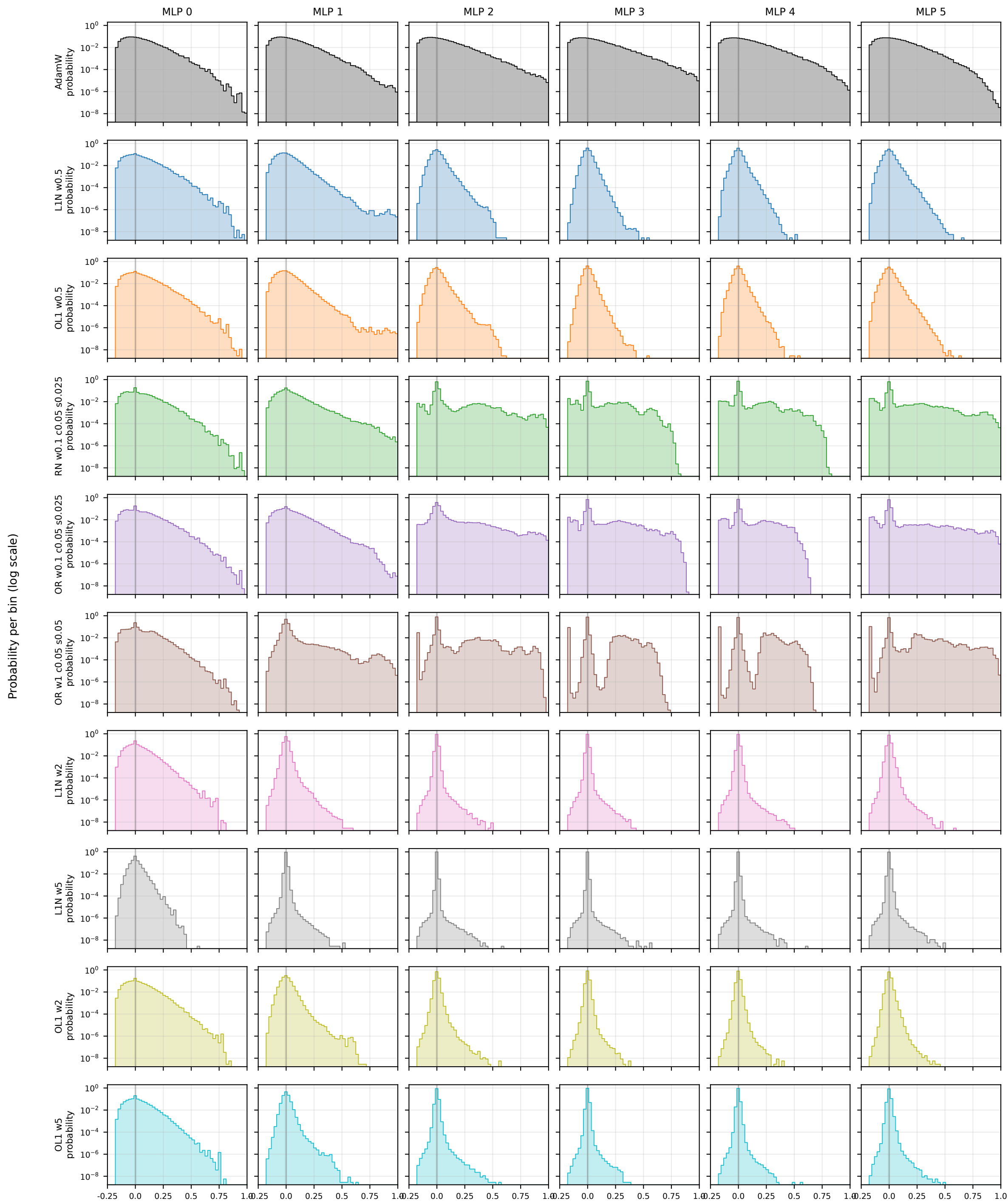
Figure 5: High-pressure OR and L1 weight-norm diagnostics. Top panels show near-zero activation mass and global weight norm trajectories; lower panels show final-checkpoint near-zero activation mass versus global and MLP-only weight norms with linear fits.

Validation Activation Histograms

The next page embeds the full validation activation histogram grid at its native figure page size. The grid compares selected AdamW, L1, Ricker, and orthogonal-pressure checkpoints on the deterministic validation pass. Each row is one method, each column is one pressure-targeted MLP activation layer, the x-axis is capped at activation value 1.0 for readability, and the y-axis is log-scaled probability per bin.

Selected Method MLP Activation Distributions

n=10 checkpoints; 338 validation blocks; 692,224 validation tokens; y-axis is log probability per bin



Shaded band marks $|activation| \leq 0.01$.